

Mohan Kumar S

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EDUCATION

SASTRA Deemed University

B.Tech in Computer Science and Engineering

CGPA: 7.66/10

Aug 2023 – Jun 2027

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL

AI / ML: PyTorch, PyTorch Geometric, Gymnasium, NumPy, Scikit-learn

Backend & Web: FastAPI, Node.js, Express.js, Next.js, REST APIs

Databases: PostgreSQL, MongoDB, MySQL, Redis, FAISS

DevOps & Cloud: Docker, GitHub Actions CI/CD, Linux, Bash

Tools: Git, Prisma ORM, Streamlit, Grafana, Cursor, Claude Code

PROJECTS

StreetPrint | Node.js, TypeScript, Express.js, MongoDB, Redis, Docker

- Built a location intelligence system that transforms raw mobility data into personalized exploration insights, enabling users to visualize travel coverage and discover unexplored areas.
- Developed a high-throughput geospatial backend leveraging MongoDB spatial indexing, Redis caching, and queue-driven ingestion pipelines for efficient processing of route telemetry.
- Engineered geospatial analytics services for route history reconstruction, heatmap generation, and location recommendation with comprehensive automated testing and containerized deployment workflows.

Sparse Mixture-of-Experts | Python, PyTorch

- Implemented a Mixture-of-Experts architecture from scratch featuring top-k noisy gating, expert routing, and capacity-aware token dispatch mechanisms.
- Matched dense-model perplexity on TinyStories while activating only a subset of parameters per token, reducing computational cost during inference.
- Analyzed routing entropy, expert utilization, and load balancing through ablation studies across multiple routing configurations.

MSGAL | PyTorch, PyTorch Geometric, GRU, GAT

- Reproduced the Multi-Granular Graph Attention Learning framework for stock movement prediction using graph-based and temporal feature extraction pipelines.
- Implemented the Joint Optimization Layer, MLP prediction module, and end-to-end training workflow integrating GRU and Graph Attention Network representations.
- Constructed correlation-based financial graphs and evaluated predictive performance on S&P 500 market data using rolling-window experimentation.

RESEARCH EXPERIENCE

FedXMSPPRA-GA | Python, SimPy, Genetic Algorithms, BFT Consensus

2025 – Present

- Designed a federated edge-fog-cloud scheduling framework for healthcare workloads using multi-objective genetic optimization across latency, energy consumption, and resource utilization objectives.
- Integrated Byzantine Fault Tolerant consensus mechanisms to improve resilience against malicious or unreliable nodes in distributed execution environments.
- Validated scheduling performance using healthcare-inspired workloads derived from MIMIC-IV and eICU datasets; manuscript in preparation.

HACKATHONS & COMPETITIONS

PharmaTriageEnv | Gymnasium, FastAPI, Docker

Meta Scaler Hackathon

- Built an OpenAI Gym-compliant multi-step RL environment for pharmacovigilance triage; modeled FAERS-inspired adverse drug event cases across 12 drugs with partial observability, DDI detection, and adversarial noise across 3 difficulty tiers.
- Designed a deterministic 5-dimension grader (severity, seriousness, expectedness, escalation, safety compound) with safety-weighted reward shaping in $[-10, +10]$.
- Dockerized the environment with CI/CD workflows targeting HuggingFace Spaces deployment and reproducible evaluation pipelines.

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI & Stanford Online (Feb 2026)
- **Docker Essentials: A Developer Introduction** – IBM Developer Skills Network (Mar 2026)